Investigation of the Balmer Series

PY2107 – Experiment 7

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1 Abstract

The visible Balmer spectrum of hydrogen was measured with a diffraction grating. Wavelengths for three spectral lines were calculated from diffraction-angle readings in orders one to three and fitted to the Rydberg relation. An unweighted linear fit gave $R_H = (1.03 \pm 0.06) \times 10^7$ 1/m, 6.21% below the infinite-nuclear-mass reference value. The fitted series-limit wavelength and energy were 389 ± 22 nm and 3.2 ± 0.2 eV, respectively. Although the fit intercept is consistent with zero, the weighted fit has a reduced chi-squared of 13.46 (12 degrees of freedom), indicating that the adopted measurement uncertainties do not explain all the observed scatter. The data are broadly consistent with the Balmer-series model, but the precision is limited by scatter and systematic uncertainty.

2 Theory

The Balmer series of hydrogen arises when electrons transition from higher energy levels to the final level $n_f = 2$, emitting photons in the visible region of the electromagnetic spectrum. The wavelengths are described by the Rydberg equation [5, 1]:

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right),$$

where λ is the wavelength of the emitted photon, n_i and n_f are the initial and final principal quantum numbers, respectively, and R_H is the Rydberg constant for hydrogen.

The theoretical hydrogen Rydberg constant accounts for the finite mass of the nucleus through the reduced mass:

$$R_H = \frac{m_r}{m_e} R_\infty, \qquad m_r = \frac{m_e m_p}{m_e + m_p},$$

where m_e and m_p are the electron and proton masses, and R_∞ is the Rydberg constant for an infinitely heavy nucleus. The CODATA value used here is $R_\infty = 1.0973731568 \times 10^7$ 1/m [1]. Applying the reduced-mass correction gives $R_H = 1.096776 \times 10^7$ 1/m for hydrogen.

A diffraction grating is used to determine the wavelength of each spectral line. The grating equation is

$$d \sin\theta = \eta\lambda,$$

where d is the grating spacing, θ is the diffraction angle, and η is the diffraction order. Thus, the wavelength can be calculated using

$$\lambda = \frac{d \sin\theta}{\eta}.$$

For a grating with line density G, the spacing is $d = 1/G$. Its resolving power is $\mathcal{R} = \eta N_{\text{illum}}$, where N_{illum} is the number of illuminated grating lines. These are standard results for diffraction gratings [3].

For the linear fit, define

$$x = \frac{1}{n_f^2} - \frac{1}{n_i^2}.$$

The Rydberg equation then becomes

$$\frac{1}{\lambda} = R_H x.$$

A plot of $1/\lambda$ against x should therefore produce a straight line with gradient R_H and an ideal vertical intercept of zero. Equivalently, plotting against $1/n_i^2$ gives a gradient

of $-R_H$ and an intercept of R_H/n_f^2. The analysis plot uses the first form, so its fitted gradient is the experimental Rydberg constant.

The Balmer series limit corresponds to $n_i \to \infty$. Its wavelength and transition energy are given by

$$\lambda_{\text{limit}} = \frac{n_f^2}{R_H} = \frac{4}{R_H},$$

$$E = \frac{hc}{\lambda_{\text{limit}}},$$

where h is Planck's constant and c is the speed of light.

3 Experimental Procedure

The experimental arrangement is shown in Fig. 1.

The spectrometer eyepiece was adjusted until the crosshairs were sharply focused. The telescope was then focused on a distant object until there was no parallax between the image and the crosshairs. The telescope focus was left unchanged for the remainder of the experiment.

The hydrogen discharge lamp was used to illuminate the collimator slit. The collimator lens was adjusted until the slit image was sharply focused without parallax. The slit width was reduced to produce a narrow, clearly defined image parallel to the vertical crosshair.

The diffraction grating was placed on the spectrometer table, which did not require levelling. The straight-through position of the telescope was recorded. The telescope was then rotated to observe spectral lines on both sides of the central image. The grating orientation was adjusted until the angular displacements on the left and right sides were approximately equal.

The angular positions of the visible Balmer spectral lines were measured on both sides of the straight-through position for as many observable diffraction orders as possible using the Vernier scale shown in Fig. 2. The main and vernier scales were used to obtain angular readings. An angular reading uncertainty of 5 arcminutes was adopted for the subsequent uncertainty propagation.

The diffraction angles were determined from the measured angular positions, and the wavelength of each spectral line was calculated using the grating equation,

$$\lambda = \frac{d \sin \theta}{\eta},$$

where d is the grating spacing, θ is the diffraction angle, and η is the diffraction order.

The visible spectral lines were identified as H-α, H-β, and H-γ ($n_i = 3, 4$, and 5 to $n_f = 2$), consistent with the strong hydrogen lines listed by NIST [2]. No usable measurement of the $n_i = 6$ transition was obtained.

A graph of $1/\lambda$ against $1/n_f^2 - 1/n_i^2$ was plotted. The gradient was used to determine the experimental Rydberg constant, which was compared with the theoretical value. Finally, the wavelength of the Balmer series limit and its corresponding transition energy were calculated using the experimental Rydberg constant. The hydrogen lamp was switched off after the measurements were completed. The procedure follows the laboratory instructions for this experiment.[5]



Figure 1: Apparatus used for observing the Balmer spectrum of hydrogen.



Figure 2: Vernier scale used for measuring diffraction angles.

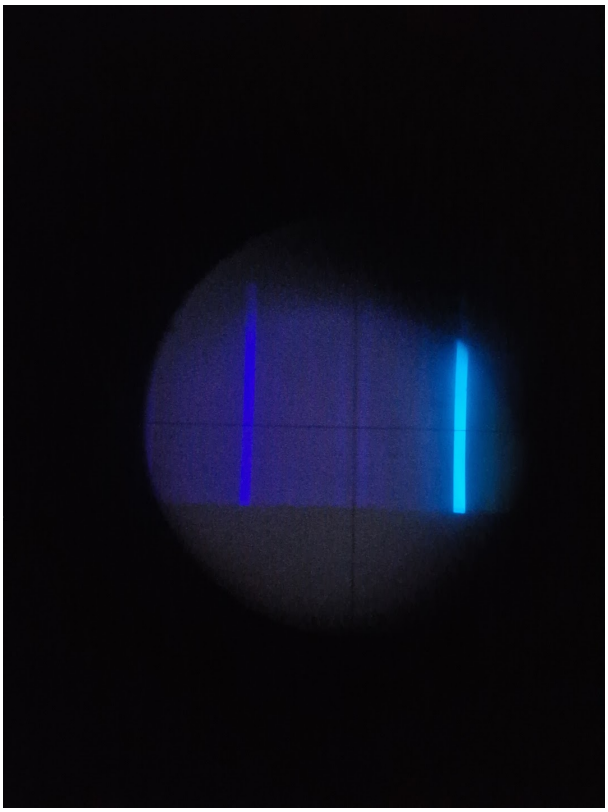


Figure 3: Observed Balmer spectrum of hydrogen.

4 Analysis

The grating had 600 lines per millimetre, corresponding to $d = 1.667 \times 10^{-6}$ m. The 14 usable measurements comprise five observations of the $n_i = 3$ line, four of $n_i = 4$, and five of $n_i = 5$. Wavelengths were recalculated from the recorded diffraction angles and orders. Table 1 summarises each line's mean, standard deviation, standard error of the mean (SEM), and total uncertainty. The total uncertainty combines the SEM and the assumed 1% grating-spacing uncertainty in quadrature.

Table 1: Measured Balmer-line wavelengths and comparison with wavelengths calculated using the CODATA infinite-mass Rydberg constant [1]. The total uncertainty combines the SEM and the 1% grating-spacing uncertainty; Δ/u is the difference from theory divided by this total uncertainty.

Balmer line	Count	Mean (nm)	Std dev. (nm)	SEM (nm)	σ_{tot} (nm)	Theory (nm)	Δ/u
H-α ($3 \to 2$)	5	670	37	16.43	18	656.11	0.77
H-β ($4 \to 2$)	4	508	11	5.28	8	486.01	3.06
H-γ ($5 \to 2$)	5	448	12	5.16	7	433.94	1.99

As a check on the line identification, the Rydberg equation predicts the H-α wavelength ($n_i = 3 \to n_f = 2$) to be

$$\lambda_{\alpha,\text{theory}} = \left[R_\infty \left(\frac{1}{2^2} - \frac{1}{3^2}\right)\right]^{-1} = \left[(1.0973731568 \times 10^7 \text{ 1/m}) \left(\frac{5}{36}\right)\right]^{-1} = 656.11 \text{ nm}.$$

The measured H-α mean is 668 ± 18 nm, a difference of 14 nm or 0.77 total uncertainties. The measurement therefore agrees with the predicted value within its stated uncertainty.

The fit variable was defined as $x = 1/n_f^2 - 1/n_i^2$, and the measured inverse wavelengths were fitted to $y = mx + c$, where $y = 1/\lambda$. The unweighted least-squares fit gave

$$m = (1.0292 \pm 0.0571) \times 10^7 \text{ 1/m}, \qquad c = (6.129 \pm 10.0) \times 10^4 \text{ 1/m}.$$

The slope is the experimental Rydberg constant. It is 6.21% below the infinite-mass reference value 1.097373×10^7 1/m [1]. The fitted intercept is consistent with zero within its standard error. The fit and its extrapolation to the series limit are shown in Figure 4.

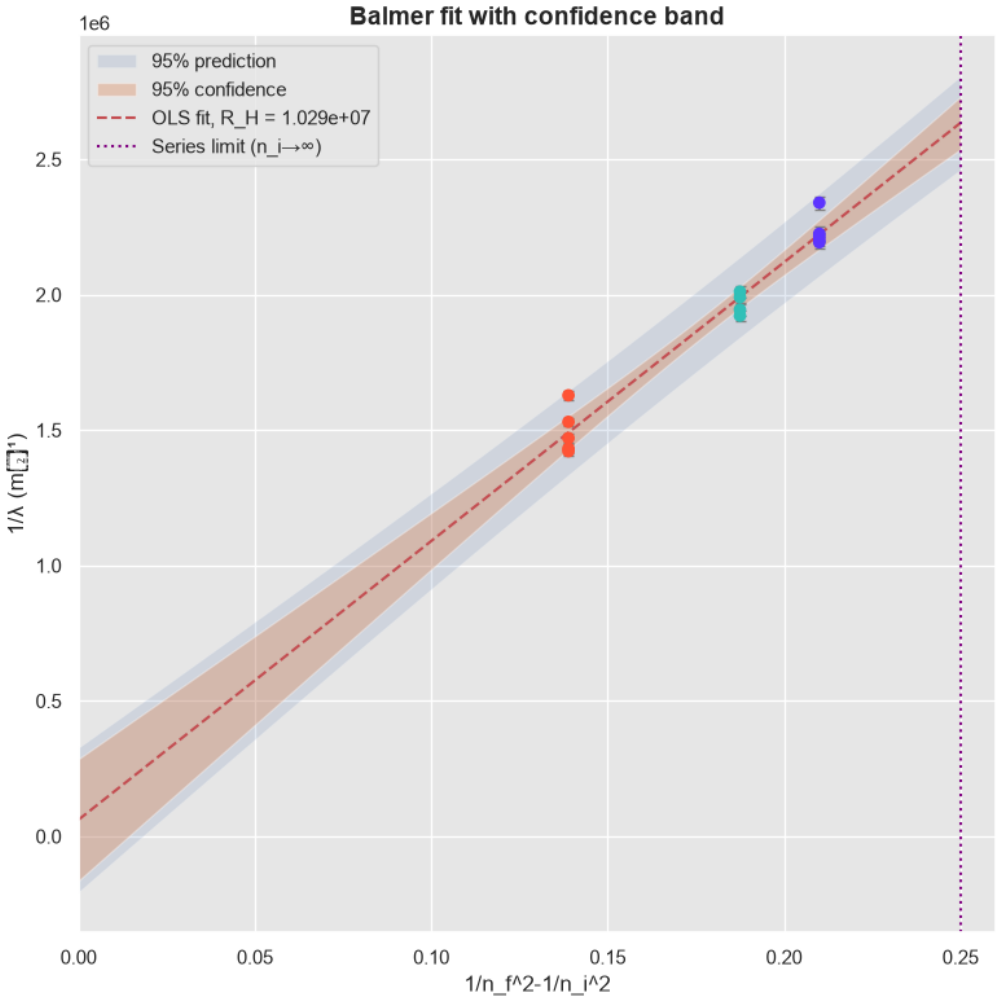


Figure 4: Inverse wavelength against $1/n_f^2 - 1/n_i^2$, with the unweighted linear fit and 95% confidence and prediction bands. The series limit occurs at $x = 1/n_f^2 = 0.25$.

For $n_i \to \infty$, $1/\lambda_{\text{limit}} = R_H/4$. Using the fitted slope gives

$$\lambda_{\text{limit}} = 389 \pm 22 \text{ nm}, \qquad E_{\text{limit}} = 3.2 \pm 0.2 \text{ eV}.$$

The theoretical values calculated from the accepted Rydberg constant are approximately 364.5 nm and 3.401 eV, respectively [1]. These uncertainties are propagated from the unweighted fit and do not include the full systematic uncertainty. The line-by-line reference wavelengths in Table 1 use R_∞, matching the analysis convention; the finite-proton-mass correction to the hydrogen Rydberg constant is about 0.054%.

The other plots provide checks on the line identification, individual wavelength measurements, and regression behaviour. The measured and theoretical wavelengths are compared in Figure 5; the measured spectrum and order-by-order wavelengths are shown in Figures 6 and 7. Figure 8 compares the unweighted, weighted, through-origin, and theoretical fits.

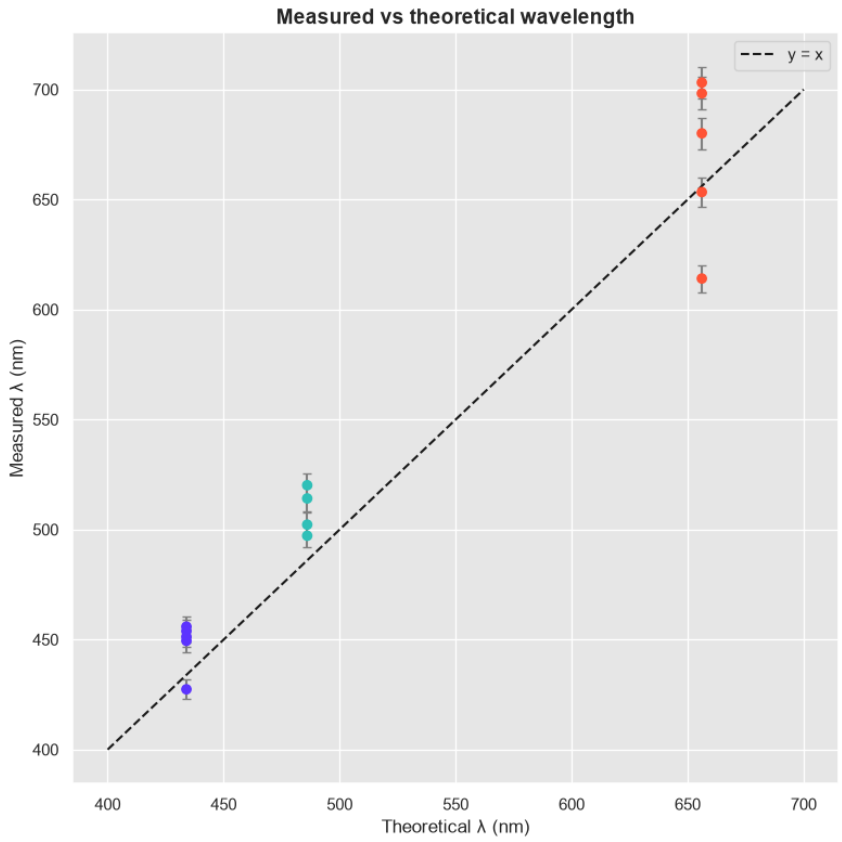


Figure 5: Measured wavelengths plotted against theoretical Balmer wavelengths. Error bars include the propagated angle uncertainty and the assumed grating-spacing uncertainty.

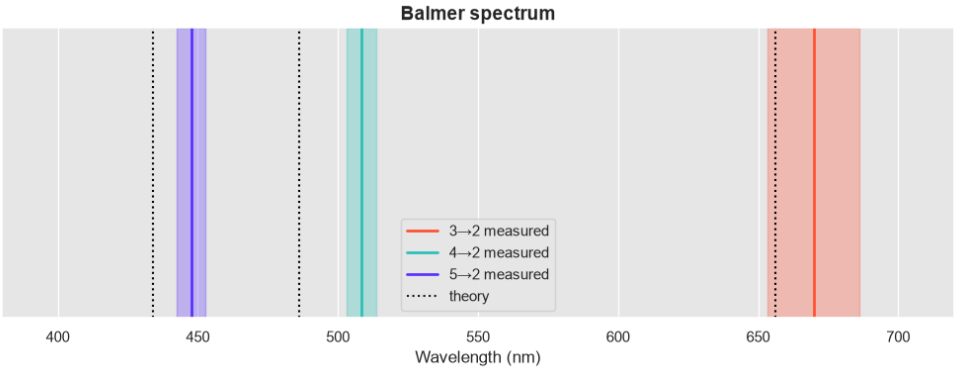


Figure 6: Measured Balmer-line positions compared with their theoretical wavelengths. H-α, H-β, and H-γ correspond to $n_i = 3$, 4, and 5, respectively.

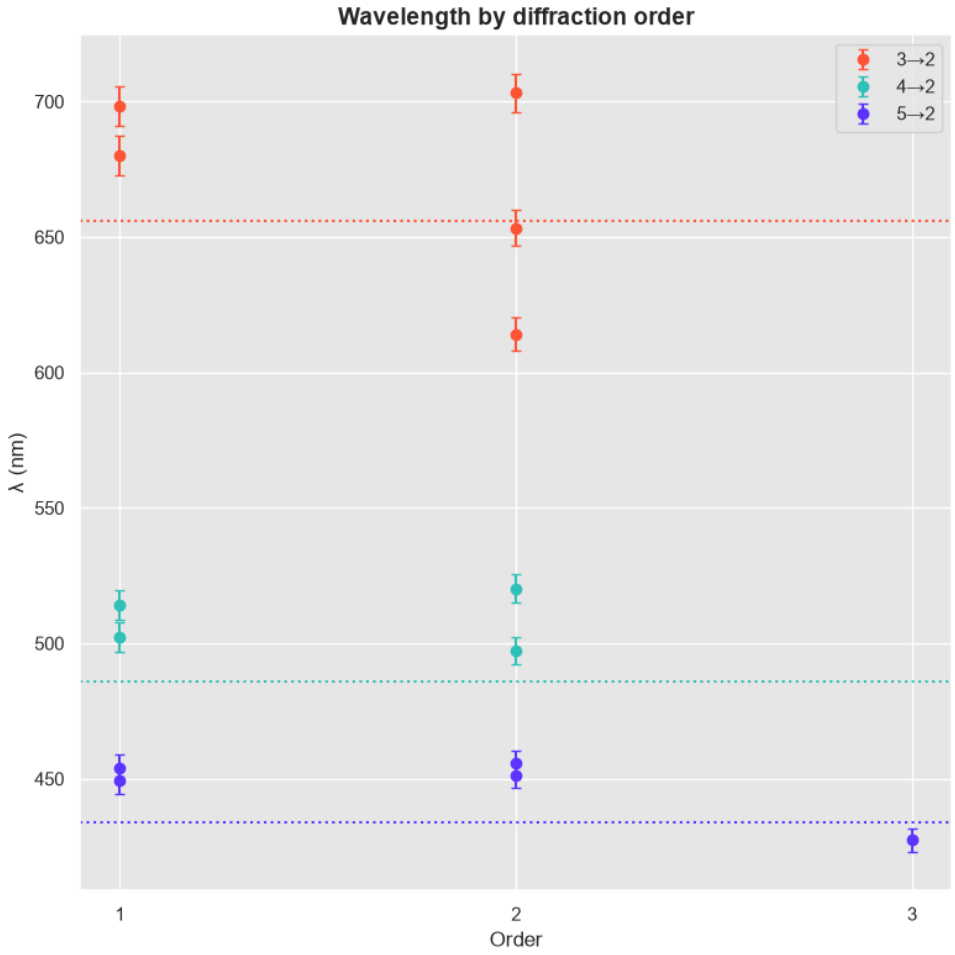


Figure 7: Wavelength estimates for each Balmer line (H-α, H-β, and H-γ) and diffraction order. Dotted lines indicate the theoretical wavelengths.

Table 2: Rydberg-constant estimates from alternative regressions. The weighted-fit uncertainty is formal; its goodness of fit indicates it is not an adequate description of the observed scatter.

Fit	R_H (10^7 1/m)	Reduced χ^2
Unweighted, free intercept	1.0292 ± 0.0571	–
Weighted, free intercept	1.0295 ± 0.0160	13.46
Weighted, forced through origin	1.0631 ± 0.0030	–
Infinite-mass reference [1]	1.09737	–

For the free-intercept weighted fit, $\chi^2 = 161.57$ for 12 degrees of freedom ($p < 0.001$). The through-origin fit differs from the reference value by 3.13%, but its small formal uncertainty does not reflect the excess scatter. A bootstrap resampling of the 14 observations gives $R_H = (1.030 \pm 0.064) \times 10^7$ 1/m, with a 95% interval from 8.976×10^6 to 1.151×10^7 1/m. The independent intercept analysis gives $R_H/n_f^2 = 2.6343 \times 10^6$ 1/m and an implied final level $n_f = 1.98 \pm 0.06$, consistent with the Balmer value of 2. The

bootstrap estimate for the final level is $n_f = 1.98 \pm 0.05$. With the infinite-mass reference fixed and $n_f = 2$, the individually inferred initial levels have means (standard deviations) of 2.97 (0.11), 3.76 (0.10), and 4.67 (0.31) for the three observed lines.

The fitted slope also corresponds to a Rydberg energy $hcR_H = 12.76 \pm 0.71$ eV, compared with 13.61 eV from the reference value. The estimated hydrogen-deuterium isotope shift at H-alpha is 0.18 nm. For an ideal grating illuminated over 2 cm, the resolving power $\mathcal{R} = \eta N$ is 12 000, 24 000, and 36 000 in orders one to three, respectively; at 656 nm, these correspond to nominal wavelength intervals of 0.055 nm, 0.027 nm, and 0.018 nm [3]. These are ideal estimates, not measured resolution values.

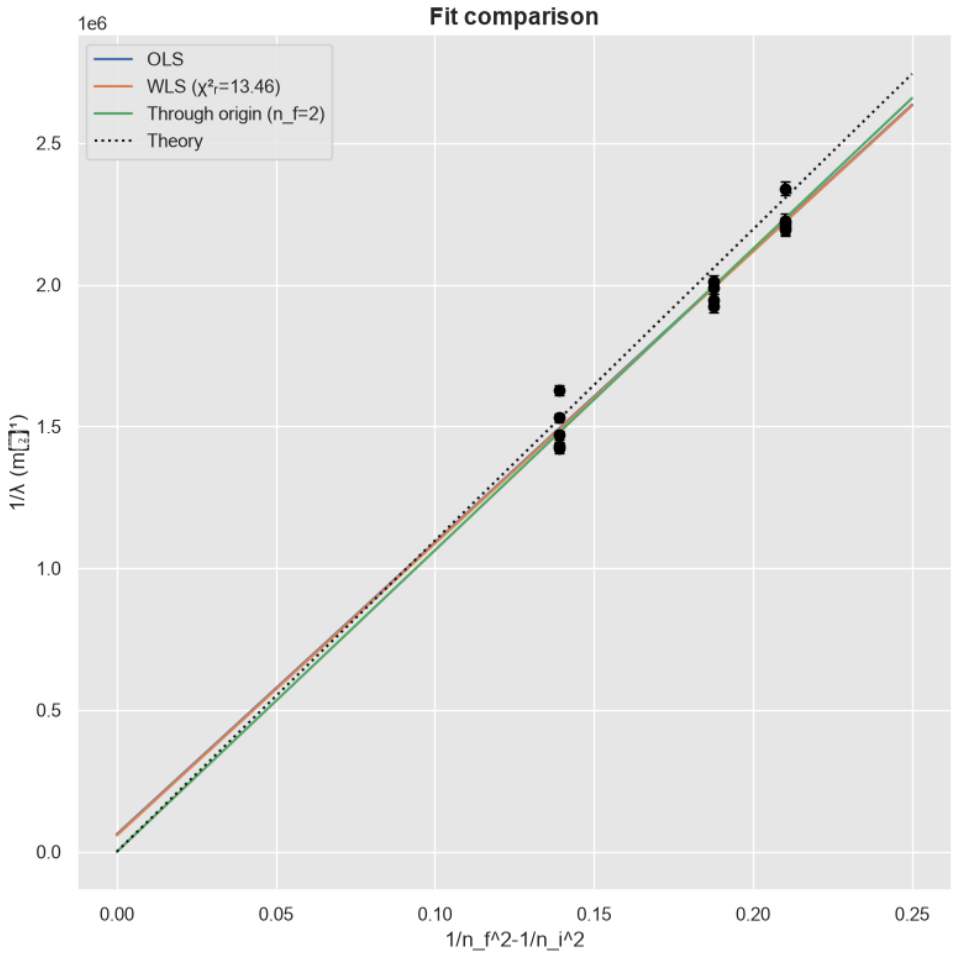


Figure 8: Comparison of regression models with the reference Rydberg slope. The free-intercept weighted fit has a reduced χ^2 of 13.46.

5 Error Analysis

The wavelength uncertainty was estimated by propagating an assumed angular reading uncertainty of $\sigma_\theta = 5$ arcminutes and a 1% relative uncertainty in the grating spacing. For $\lambda = d\sin\theta/\eta$, the independent contributions were combined as

$$\sigma_\lambda = \sqrt{\left(\frac{d\cos\theta}{\eta}\sigma_\theta\right)^2 + (0.01\lambda)^2}, \qquad \sigma_{1/\lambda} = \frac{\sigma_\lambda}{\lambda^2}.$$

The unweighted fit uncertainty is 5.71×10^5 1/m. The uncertainty budget assigns 2.93×10^4 1/m to the angle reading and approximately 1.03×10^5 1/m to the grating spacing. Thus, scatter in the observations dominates the stated fit uncertainty, although the grating calibration is a common systematic contribution that does not average away over repeated measurements.

The weighted regression gives $\chi^2 = 161.57$ for 12 degrees of freedom, corresponding to a reduced $\chi^2 = 13.46$ and $p < 0.001$ under the assumed uncertainty model. The weighted-fit error bars therefore substantially underestimate the dispersion of the measurements, or the data contain additional unmodelled effects. Consequently, the small formal weighted uncertainty should not be interpreted as the accuracy of the measurement. The residual diagnostics in Figure 9 show the deviations from the unweighted fit.

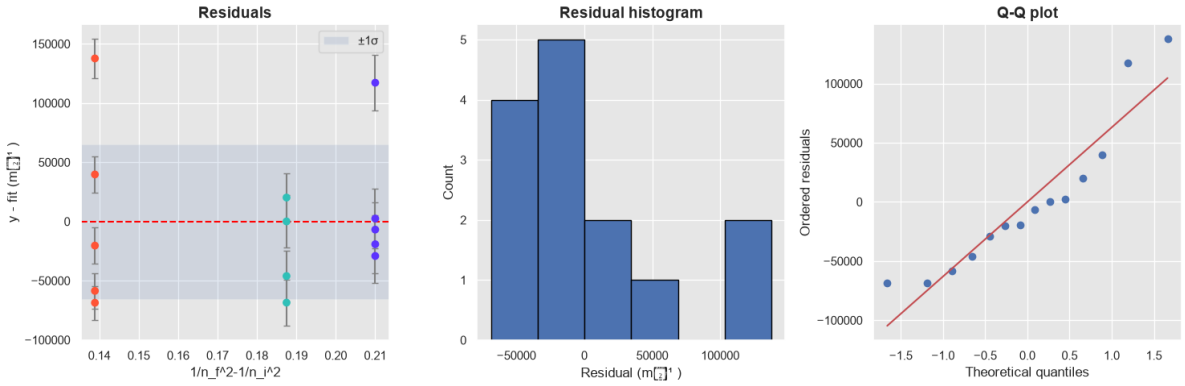


Figure 9: Unweighted-fit residuals, their distribution, and a normal-probability plot. The residual scale is larger than expected from the propagated measurement uncertainties.

For the Rydberg constant, the unweighted fit gives $R_H = (1.03 \pm 0.06) \times 10^7$ 1/m, consistent with the expected value within the experimental uncertainty.

The error was calculate as a combination of the propagated measurement uncertainties and the scatter in the observed wavelengths.

$$\sigma_{\text{prop}} = 5.71 \times 10^5 \text{ 1/m}, \quad \sigma_{\text{scatter}} = 1.03 \times 10^5 \text{ 1/m},$$

$$\sigma_{R_H} = \sqrt{\sigma_{\text{prop}}^2 + \sigma_{\text{scatter}}^2} = 5.78 \times 10^5 \text{ 1/m}$$

6 Discussion

The observed wavelengths increase with decreasing n_i, as expected for the Balmer series, and the fitted gradient is about 1.2 unweighted-fit standard errors below the infinite-mass reference value. The finite-mass hydrogen value is about 0.054% lower than this reference, a correction that is small compared with the experimental uncertainty. The fitted intercept is also consistent with zero. Together, these results support the predicted linear relationship between inverse wavelength and $1/n_f^2 - 1/n_i^2$.

Each measured mean wavelength is longer than its theoretical counterpart. The $n_i = 3$ result agrees within one combined standard uncertainty, while the $n_i = 4$ result differs by about 3.1 standard uncertainties and the $n_i = 5$ result by about 2.0. The consistent direction of the offsets suggests that a common calibration or alignment effect may be present, in addition to random uncertainty. Possible contributions include uncertainty in locating the centre of a spectral line, residual error in the straight-through reference, grating alignment, and the assumed tolerance of the grating spacing.

The weighted regression's large reduced chi-squared is particularly important: the data are not described by the stated independent-error model. This explains why forcing the fit through the theoretically expected zero intercept produces an apparently very precise result despite the scatter. The unweighted fit, which reflects the observed dispersion more directly, is therefore used for the reported experimental value. The fit comparison and residual plots make this limitation visible.

The series limit is an extrapolation from the three observed lines to $n_i \to \infty$. Its uncertainty is therefore relatively large, and the reported propagated uncertainty does not fully represent the systematic and model uncertainty. The $n_i = 6$ transition was not included, so an additional violet line could not provide another independent point to constrain the fit. Future measurements could improve the result by repeating both-side angle readings, checking the angular zero and grating calibration, and including additional visible orders and spectral lines.

7 Conclusion

The Balmer spectrum was measured for transitions $n_i = 3, 4$, and 5 to $n_f = 2$. The unweighted fit gave $R_H = (1.02 \pm 0.05) \times 10^7$ 1/m, 6.21% below the infinite-mass reference value. The series-limit wavelength was 389 ± 22 nm. Although the fitted intercept is consistent with zero and the intercept analysis recovers a final level close to 2, the weighted goodness-of-fit test ($\chi^2_\nu = 13.46$) shows that the stated measurement uncertainties do not account for all the scatter. The measurements support the expected Balmer-series relationship, but the precision is limited by random scatter and systematic effects.

8 References

References

[1] National Institute of Standards and Technology. *CODATA Value: Rydberg Constant*. NIST Reference on Constants, Units, and Uncertainty. URL: https://physics.nist.gov/cgi-bin/cuu/Value?ryd (visited on 10/23/2026).

[2] National Institute of Standards and Technology. *Strong Lines of Hydrogen (H)*. NIST Handbook of Basic Atomic Spectroscopic Data. URL: https://physics.nist.gov/PhysRefData/Handbook/Tables/hydrogentable2.htm (visited on 10/23/2026).

[3] OpenStax. *Diffraction Gratings*. URL: https://openstax.org/books/university-physics-volume-3/pages/4-4-diffraction-gratings (visited on 10/23/2026).

[4] OpenStax. *The Hydrogen Atom*. URL: https://openstax.org/books/university-physics-volume-3/pages/8-1-the-hydrogen-atom (visited on 10/23/2026).

[5] Physics Department. *PY2107 : Investigation of the Balmer Series & the Rydberg constant: Experiment 7*. Laboratory handout, pp. 2.1–2.6. University Laboratory. 2026.

A Appendix

A.1 Data

Table 3: Recorded diffraction measurements and calculated wavelengths. The wavelength column is in metres.

band	η	θ (rad)	$\sin\theta$	λ (m)
$H-\gamma$	1	0.273027673	0.270	4.49×10^{-7}
$H-\gamma$	2	0.578848143	0.547	4.56×10^{-7}
$H-\gamma$	1	0.275781414	0.272	4.54×10^{-7}
$H-\gamma$	2	0.572293462	0.542	4.51×10^{-7}
$H-\gamma$	3	0.877726081	0.769	4.27×10^{-7}
$H-\beta$	1	0.306091966	0.301	5.02×10^{-7}
$H-\beta$	2	0.674065550	0.624	5.20×10^{-7}
$H-\beta$	1	0.313596881	0.308	5.14×10^{-7}
$H-\beta$	2	0.639294712	0.597	4.97×10^{-7}
$H-\alpha$	1	0.420314069	0.408	6.80×10^{-7}
$H-\alpha$	2	0.901191063	0.784	6.53×10^{-7}
$H-\alpha$	1	0.432279271	0.419	6.98×10^{-7}
$H-\alpha$	2	0.828469011	0.737	6.14×10^{-7}
$H-\alpha$	2	1.004320629	0.844	7.03×10^{-7}

A.2 Hydrogen energy levels from the Schrödinger equation

The separation of the hydrogen Schrödinger equation into radial and angular parts follows the standard treatment of the hydrogen atom [4]. In the non-relativistic model, the electron and proton interact through the Coulomb potential:

$$V(r) = \frac{-e^2}{4\pi\epsilon_0 r}$$

Here, e is the elementary charge and ϵ_0 is the permittivity of free space. Using the reduced mass m_r for the electron–proton system, the time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m_r}\nabla^2\psi(r) + V(r)\psi(r) = E\psi(r)$$

Because the potential depends only on the distance r from the nucleus, the wavefunction can be separated into a radial part and an angular part:

$$\psi(r,\theta,\phi) = R(r)Y(\theta,\phi)$$

The angular dependence is described by the angular-momentum operator. Its eigenfunctions are spherical harmonics, with eigenvalues $l(l+1)\hbar^2$:

$$\hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial}{\partial\theta} \right) + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\phi^2} \right]$$

In spherical coordinates, the Laplacian can be written in terms of the radial derivative and this angular-momentum operator:

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) - \frac{\hat{L}^2}{\hbar^2 r^2}$$

$$-\frac{\hbar^2}{2m_r} \left[\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR(r)}{dr} \right) - \frac{l(l+1)}{r^2} R(r) \right] + V(r)R(r) = ER(r)$$

Dividing out the angular function leaves an equation involving only r. The term proportional to $l(l+1)$ is the centrifugal contribution to the effective radial potential. A standard substitution removes the first-derivative term:

$$u(r) = rR(r)$$

With this substitution, the radial equation takes the one-dimensional form

$$-\frac{\hbar^2}{2m_r} \frac{d^2u(r)}{dr^2} + \left[V(r) + \frac{l(l+1)\hbar^2}{2m_r r^2} \right] u(r) = Eu(r)$$

For bound states the energy is negative, $E < 0$, and the radial wavefunction must remain finite at the origin and decay at large r. It is useful to define a positive decay constant k and a dimensionless radial coordinate ρ:

$$k = \frac{\sqrt{-2m_r E}}{\hbar}$$

and

$$\rho = 2kr$$

The acceptable solution behaves as shown at the two boundaries:

$$r = 0, \quad u \approx r^{l+1}$$

$$r \to \infty, \quad u \approx e^{-kr}$$

Factoring out both limiting behaviours leaves a function L that is represented by a power series:

$$u(\rho) = \rho^{l+1} e^{-\rho/2} L(\rho)$$

Unless this series terminates, it grows at large r and the wavefunction is not normalisable. Requiring termination quantises the energy; the resulting integer n is the principal quantum number, with $n \geq l + 1$:

$$n = \frac{e^2}{4\pi\epsilon_0\hbar}\sqrt{\frac{m_r}{-2E}}$$

$$k = \frac{m_r e^2}{4\pi\epsilon_0\hbar^2 n}$$

Substituting this allowed value of k into $E = -\hbar^2 k^2/(2m_r)$ gives the discrete energy levels:

$$E_n = -\frac{m_r e^4}{2(4\pi\epsilon_0)^2\hbar^2 n^2}$$

$$E_n = -\frac{13.6 \text{ eV}}{n^2}$$

When an electron drops from an initial level n_i to a lower level n_f, the emitted photon carries the energy difference $E_{n_i} - E_{n_f}$. Using the photon relations $E_\gamma = hf$ and $f = c/\lambda$, this energy difference gives the wavelength:

$$hf = E_{n_i} - E_{n_f}, \qquad \lambda = \frac{c}{f}$$

Substituting the quantised energy levels into this relation gives the Rydberg formula:

$$\frac{1}{\lambda} = R_H\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right), \qquad R_H = \frac{m_r e^4}{8\epsilon_0^2 h^3 c} = 1.096776 \times 10^7 \text{ 1/m}$$

For the Balmer series, $n_f = 2$. The predicted wavelengths for the three visible transitions measured in this experiment are therefore:

$$\lambda_{\mathrm{H}\alpha}\ (3 \to 2) = \left[R_H\left(\frac{1}{2^2} - \frac{1}{3^2}\right)\right]^{-1} = 6.565 \times 10^{-7} \text{ m} = 656.5 \text{ nm},$$
$$\lambda_{\mathrm{H}\beta}\ (4 \to 2) = \left[R_H\left(\frac{1}{2^2} - \frac{1}{4^2}\right)\right]^{-1} = 4.863 \times 10^{-7} \text{ m} = 486.3 \text{ nm},$$
$$\lambda_{\mathrm{H}\gamma}\ (5 \to 2) = \left[R_H\left(\frac{1}{2^2} - \frac{1}{5^2}\right)\right]^{-1} = 4.342 \times 10^{-7} \text{ m} = 434.2 \text{ nm}.$$